A Comparative Evaluation of Socratic Multi-Agent Dialogue in AI Tutoring

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Abstract

We evaluate conversational learning efficiency using two-host Socratic dialogue versus traditional didactic monologue across 1,240 secondary school participants. Empirical findings demonstrate a 34% increase in concept retention.

Methodology and Empirical Cohorts

Cohort	Modality	Participants	Retention Rate (14 Days)
Group A	Static Textbook	410	51.4%
Group B	Unimodal Monologue	415	62.8%
Group C	Socratic 2-Host Dialogue	415	85.4%

Results & Discussion

Statistical significance confirmed via two-tailed t-test ($t = 8.42$, $p < 0.001$).